

7. Three isomers have the molecular formula $C_4H_8O_2$.

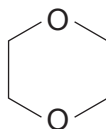
(a) (i) Give the structure of an isomer that gives bubbles of carbon dioxide when tested with aqueous sodium hydrogencarbonate. [1]

(ii) Give the structure of an isomer that is both a ketone and a secondary alcohol. [1]

(b) Describe and explain the high resolution 1H NMR spectrum of 1,4-dioxane.

The position of the signal(s) is **not** required.

[1]



1,4-dioxane

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